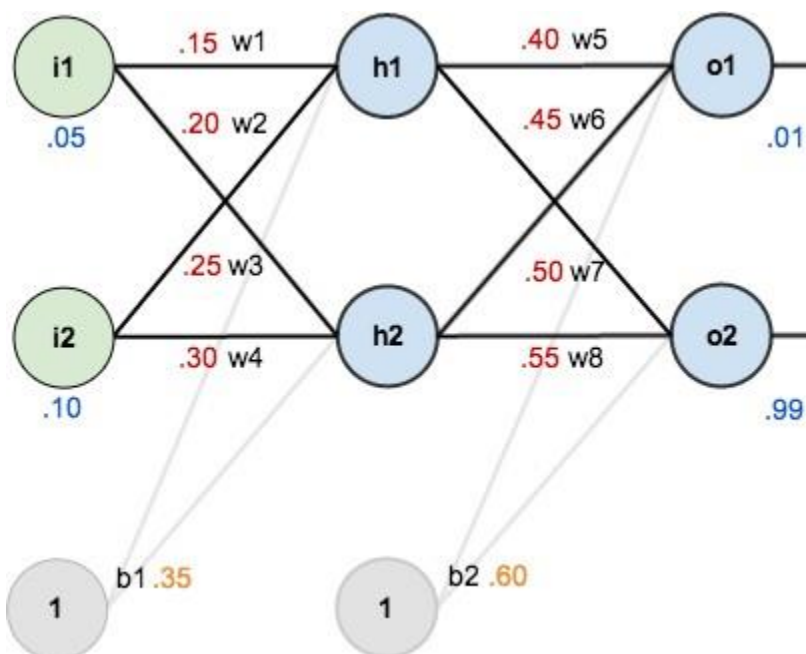




Backpropagation

Perform 1 backpropagation iteration for the following network:

- Use Sigmoid as the activation function ($Sigmoid(x) = \frac{1}{1+e^{-x}}$)
- Inputs and outputs are written in blue
- Initial weights are written in red
- Initial bias are written in yellow
- Learning rate $\eta = 0.5$



Answer:

Forward pass:

$$H_1 = x_1 * w_1 + x_2 * w_2 + b_1$$

$$H_1 = 0.05 * 0.15 + 0.1 * 0.2 + 0.35 = 0.3775$$

$$Out_{H_1} = \text{sigmoid}(H_1) = 0.5933$$

$$H_2 = x_1 * w_3 + x_2 * w_4 + b_1$$

$$H_2 = 0.05 * 0.25 + 0.1 * 0.3 + 0.35 = 0.3925$$

$$Out_{H_2} = \text{sigmoid}(H_2) = 0.5969$$

$$O_1 = Out_{H_1} * w_5 + Out_{H_2} * w_6 + b_2$$

$$O_1 = 0.5933 * 0.4 + 0.5969 * 0.45 + 0.6 = 1.1059$$

$$Out_{O_1} = \text{sigmoid}(O_1) = 0.7514$$

$$O_2 = Out_{H_1} * w_7 + Out_{H_2} * w_8 + b_2$$

$$O_2 = 0.5933 * 0.5 + 0.5969 * 0.55 + 0.6 = 1.2249$$

$$Out_{O_2} = \text{sigmoid}(O_2) = 0.7729$$

Calculating the error: (MSE)

$$E_{Total} = E_{Out_1} + E_{Out_2}$$

$$E_{Out_1} = \frac{1}{2} (0.01 - 0.7514)^2 = 0.2748$$

$$E_{Out_2} = \frac{1}{2} (0.99 - 0.7729)^2 = 0.0236$$

$$E_{Total} = 0.2748 + 0.0236 = 0.2984$$

Backward pass:

Computing gradients:

1) For the output layer:

$$\frac{\partial E_{total}}{\partial w_5} = \frac{\partial E_{total}}{\partial Out_{O_1}} * \frac{\partial Out_{O_1}}{\partial O_1} * \frac{\partial O_1}{\partial w_5}$$

$$\frac{\partial E_{total}}{\partial Out_{O_1}} = -(target_1 - Out_{O_1}) = -(0.01 - 0.7514) = 0.7414$$

$$\frac{\partial Out_{O_1}}{\partial O_1} = Out_{O_1} (1 - Out_{O_1}) = 0.7514(1 - 0.7514) = 0.1868$$

$$\frac{\partial Q_1}{\partial w_5} = Out_{H_1} = 0.5933$$

Finally, putting everything together:

$$\frac{\partial E_{total}}{\partial w_5} = 0.7414 * 0.1868 * 0.5933 = 0.0822$$

$$w_5 = w_5 - \eta * \frac{\partial E_{total}}{\partial w_5} = 0.4 - 0.5 * 0.0822 = 0.3589$$

Similarly for all weights in the output layer we get:

$$w_6 = 0.4087$$

$$w_7 = 0.5113$$

$$w_8 = 0.5614$$

2) For the hidden layer:

$$\frac{\partial E_{total}}{\partial w_1} = \frac{\partial E_{total}}{\partial Out_{H_1}} * \frac{\partial Out_{H_1}}{\partial H_1} * \frac{\partial H_1}{\partial w_1}$$

First term is calculated by adding the effect of Out_{H_1} on both E_{Out_1} and E_{Out_2} :

$$\frac{\partial E_{total}}{\partial Out_{H_1}} = \frac{\partial E_{Out_{O_1}}}{\partial Out_{H_1}} + \frac{\partial E_{Out_{O_2}}}{\partial Out_{H_1}}$$

$$\frac{\partial E_{Out_{O_1}}}{\partial Out_{H_1}} = \frac{\partial E_{Out_{O_1}}}{\partial Out_{O_1}} * \frac{\partial Out_{O_1}}{\partial O_1} * \frac{\partial O_1}{\partial Out_{H_1}}$$

$$\frac{\partial E_{Out_1}}{\partial Out_{O_1}} = -(target_1 - Out_{O_1}) = -(0.01 - 0.7514) = 0.7414$$

$$\frac{\partial Out_{O_1}}{\partial O_1} = Out_{O_1} (1 - Out_{O_1}) = 0.7514(1 - 0.7514) = 0.1868$$

$$\frac{\partial Q_1}{\partial Out_{H_1}} = w_5 = 0.40$$

$$\frac{\partial E_{Out_1}}{\partial Out_{H_1}} = 0.7414 * 0.1868 * 0.40 = 0.0554$$

Similarly for $\frac{\partial E_{Out_{O_2}}}{\partial Out_{H_1}}$:

$$\frac{\partial E_{out_2}}{\partial Out_{H_1}} = \frac{\partial E_{out_2}}{\partial Out_{O_2}} * \frac{\partial Out_{O_2}}{\partial O_2} * \frac{\partial O_2}{\partial Out_{H_1}}$$

$$\text{We will get } \frac{\partial E_{out_2}}{\partial Out_{H_1}} = -0.0191$$

$$\text{Adding this up will give us } \frac{\partial E_{total}}{\partial Out_{H_1}} = 0.0554 - 0.0191 = 0.0363$$

Now we calculate the remaining terms:

$$\frac{\partial Out_{H_1}}{\partial H_1} = Out_{H_1} (1 - Out_{H_1}) = 0.5933(1 - 0.5933) = 0.2413$$

$$\frac{\partial H_1}{\partial w_1} = x_1 = 0.05$$

Putting everything together, we get:

$$\frac{\partial E_{total}}{\partial w_1} = 0.0363 * 0.2413 * 0.05 = 0.0004$$

$$w_1 = w_1 - \eta * \frac{\partial E_{total}}{\partial w_1} = 0.15 - 0.5 * 0.0004 = 0.1498$$

Similarly for all weights in the hidden layer we get:

$$w_2 = 0.1996$$

$$w_3 = 0.2498$$

$$w_4 = 0.2995$$

Finally, we've updated all of our weights! When we fed forward the 0.05 and 0.1 inputs originally, the error on the network was 0.2984. After this first round of backpropagation, the total error is now down to 0.2910.